

TODO

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Abstract

1 Introduction

2 Selected Data Sources

3 Data Profiling

This section of the report discusses the data profiling part of the project. Section 3.1 covers the performed single column profiling steps, while Section 3.2, Section 3.3, and Section 3.4 will cover respectively the application of unique column combination, functional dependency, and inclusion dependency discovery. Finally, Section 3.5 covers the relational database schema designed for the data and how the profiling result affected design choices.

This part of the project made extensive use of the Metanome data profiling platform [1]. Each section will detail which Metanome algorithms were used. In addition to Metanome, also the YData Profiling Python library [2] has been used for single column analysis.

3.1 Single Column Analysis

Single column analysis has been performed for this project using both the YData Profiling Python library and the SCDP algorithm implementation for Metanome. The goal of this analysis is primarily to understand the type and distribution of data contained within the individual columns. As such it will influence for example the data type chosen for the corresponding attributes in the relational schema and ontology developed in Section 3.5 and Section 4.1. Below I include, in addition to some commentary about the results, also tables that include the most relevant information for each column of the various tables.

3.1.1 bz_infrastructure_line and bz_infrastructure_node

Table 1 shows the most important single column analysis results for each column of the bz_infrastructure_line table. The table contains a total of 47551 rows. We can operate that the table contains three columns that individually for a unique key, ogc_fid, glm_id, and objectid. ogc_fid is a sequentially increasing integer, going from 0 to 47551. Interestingly, objectid is stored as a real data type, with values such as "94188705.0", "94188707.0", or "94188708.0". However, all values are whole integers. The values of glm_id all follow the pattern UrbanPlan-InfrastructurePlan-Network.<objectid> where <objectid> is the integer value stored in the objectid column of the same tuple.

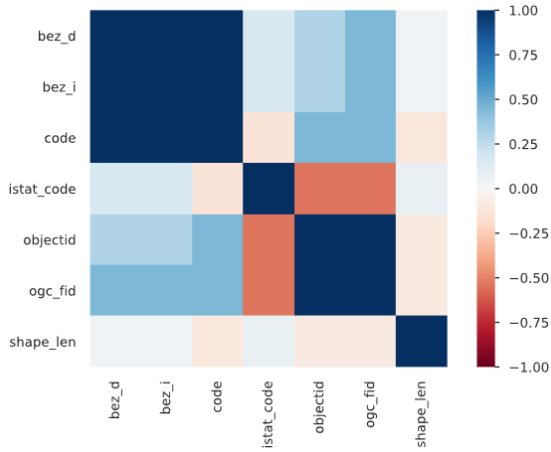
istat_code, nda_link_de, nda_link_it, gem_text_d, and gem_text_i all have 115 distinct values, indicating that they might contain related values. Manually inspecting gem_text_d reveals that it contains the names of municipalities, presumably the one the infrastructure line belongs to. gem_text_i contains the Italian name and istat_code some kind of identifier. nda_link_de and nda_link_it contain links that when followed lead to the Implementing rules to the land-use plan of the given municipality.

code, bez_i, and bez_d all have 11 distinct values. These values seem to indicate the type of infrastructure that is represented by this tuple, with the values in bez_d being the German name, those in bez_i the Italian name, and the values in code being an integer identifier of the type. Note that like with objectid, the values in code are represented as real numbers ending with ".0", meaning they all have integer values. The following are all possible values of bez_d: "Trinkwasserleitung", "Druckrohrleitung", "Abwasserleitung", "Schlammleitung", "Methangasleitung", "Hochspannungsleitung", "Mittelspannungsleitung", "Fernwärmeleitung", "Hochspannungsleitung - Bannstreifen", "Sonstige Leitungen", "Methangasleitung - Bannstreifen".

Finally, geom contains the geometry of the infrastructure line. Manual analysis reveals that the values are in the Well-Known Text format. shape_len contains codealt and blp only contain missing values

Column	Data Type	% NULLs	# distinct	Min	Max	Avg	Std.
ogc_fid	INT	0	47551	1	47551	23776	13726.94
gml_id	VARCHAR[48]	0	47551	-	-	-	-
objectid	REAL	0	47551	94188705	94462179	94295324.76	102812.43
code	REAL	0	11	1021101	1021111	1021103.63	2.42
codealt	NA	100	0	-	-	-	-
bez_i	VARCHAR[48]	0	11	-	-	-	-
bez_d	VARCHAR[48]	0	11	-	-	-	-
istat_code	REAL	0	115	21001	21118	21044.08	34.64
nda_link_de	VARCHAR[64]	0	115	-	-	-	-
nda_link_it	VARCHAR[64]	0	115	-	-	-	-
nda_text_de	VARCHAR[16]	0	1	-	-	-	-
nda_text_it	VARCHAR[16]	0	1	-	-	-	-
gem_text_d	VARCHAR[32]	0	115	-	-	-	-
gem_text_i	VARCHAR[32]	0	115	-	-	-	-
blp	NA	100	0	-	-	-	-
shape_len	REAL	0	45404	0.004	23001.3641	223.08	503.35
geom	TEXT	0	47115	-	-	-	-

Table 1: Single column analysis results for the different columns of the bz_infrastructure_line table extracted using the Metanome SCDP algorithm.



(a) Correlation matrix for bz_infrastructure_line.

(b) Correlation matrix for bz_infrastructure_node.

Figure 1: Column correlation matrix showing the correlation between all pairs of columns for the bz_infrastructure_line and bz_infrastructure_node tables. Spearman correlation is used for comparing numerical variables and Cramer’s V association coefficient for categorical variables.

3.1.2 bz_hazard_landslide and bz_hazard_avalanche

3.1.3 bz_admin_municipality

3.2 Unique Column Combination Discovery

3.2.1 bz_infrastructure_line and bz_infrastructure_node

3.2.2 bz_hazard_landslide and bz_hazard_avalanche

3.2.3 bz_admin_municipality

3.3 Functional Dependency Discovery

3.3.1 bz_infrastructure_line and bz_infrastructure_node

3.3.2 bz_hazard_landslide and bz_hazard_avalanche

3.3.3 bz_admin_municipality

3.4 Inclusion Dependency Discovery 2

3.5 Relational Database Schema

4 Data Integration

- [2] YData Labs Inc. *YData Profiling Documentation*. <https://docs.profiling.ydata.ai/latest/>. Accessed: 2026-02-10. 2026.